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AutoGRN: An Automated Graph Neural Network Framework for Gene Regulatory Network Inference

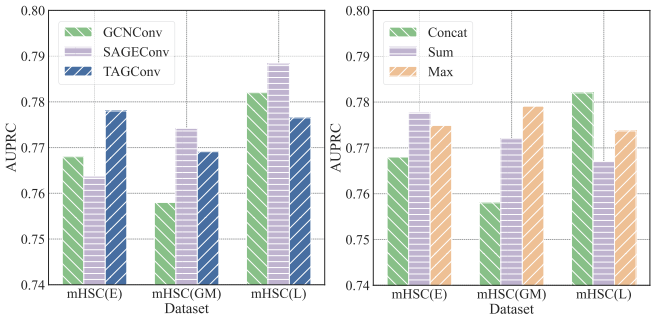
Jianliang Gao, Shuqing Wu, Siyang Xiao, and Zhenpeng Wu*

***Abstract*— Gene regulatory network (GRN) inference is essential for understanding gene interactions that control biological functions and disease progression. Its goal is to uncover regulatory relationships among genes using gene expression data, such as single-cell RNA sequencing (scRNA-seq). With the rise of deep learning, numerous approaches—particularly those using Graph Neural Networks (GNNs)—have shown promising progress in GRN inference. However, scRNA-seq datasets exhibit high sparsity, noise, dropout events, and cellular heterogeneity. These characteristics make it difficult for fixed GNN architectures to generalize across scRNA-seq datasets, often leading to performance degradation. To address this challenge, we propose AutoGRN, an automated graph neural network framework for gene regulatory network inference tasks. AutoGRN incorporates a range of components that potentially influence GRN inference performance into its search space. Through a genetic search algorithm constrained by information entropy, AutoGRN identifies optimal GNN architectures by adapting to the characteristics of individual datasets. Experimental results show that AutoGRN outperforms existing methods in prediction accuracy across multiple public datasets.**

***Index Terms*— Gene regulatory network, graph neural networks, graph neural architecture search, scRNA-seq.**

I. Introduction

GENE regulatory networks (GRNs) form the foundation for understanding the complex interactions between transcription factors (TFs) and their target genes [1], which drive processes such as cell differentiation, development, and disease progression [2]. The advancement of single-cell RNA sequencing (scRNA-seq) technology has greatly facilitated the exploration of GRNs. Compared to bulk RNA sequencing [3], which treats all cells in a sample as a whole to study gene expression, scRNA-seq technology can discern gene expression at the single-cell level, generating a two-dimensional matrix of gene expression values across cells. This enables high-resolution investigation of the hidden dynamic mechanisms underlying gene regulation at the single-cell level [4]. However, the advantages of scRNA-seq also bring new challenges. Compared to bulk RNA sequencing, scRNA-seq datasets often exhibit high sparsity, elevated noise levels, pronounced dropouts, as well as marked cellular heterogeneity [5]. Cellular heterogeneity refers to the biological differences and diversity among cells within a sample, even when they originate from the same tissue of the same organism [6]. In scRNA-seq datasets, this heterogeneity manifests as variations in the distribution patterns of gene expression profiles [7], [8]. This combination makes it difficult for fixed model architectures to generalize across diverse datasets.



(a) Convolution Type (b) Fusion Type

Fig. 1: The comparative results of AUPRC performance across (a) different convolution types and (b) feature fusion types on three distinct scRNA-seq datasets.

Traditional GRN inference methods often rely on information theory-based approaches. For example, [9] and [10] assume a linear relationship between the strength of gene correlation and the likelihood of regulatory interaction and use Pearson Correlation Coefficients (PCC) to infer regulatory interactions from data. Nonlinear approaches, such as ARACNE [11], utilize Mutual Information (MI) to capture more complex interactions beyond simple correlation by analyzing pairwise gene expression profiles. With the emergence of deep learning, methods leveraging its ability to handle high-dimensional datasets and model complex nonlinear relationships have been increasingly explored. Representative works include CNNC [12] and STGRNS [13]. Among these, Graph Neural Networks (GNNs), which represent regulatory interactions through graph structures, have demonstrated great potential for modeling local dependencies and global topologi-

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The code is available at: https://github.com/surqing/AutoGRN.

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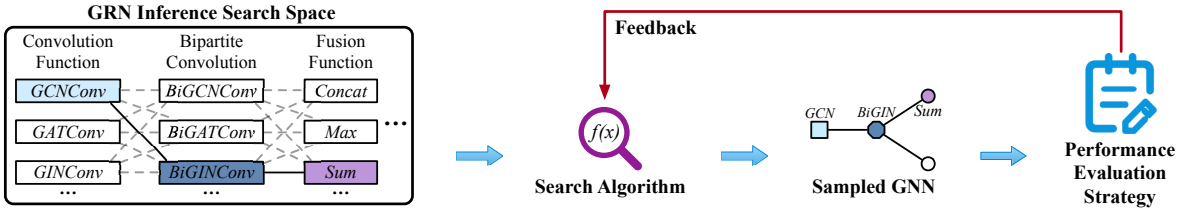


Fig. 2: The illustration of the GNAS process for gene regulatory network inference.

cal properties. GNN-based methods are increasingly applied to GRN inference, such as scMGATGRN [14], which effectively aggregates local and higher-order neighbor features extracted by multi-view graph attention networks to improve accuracy; GCLink [15], which integrates graph attention networks with graph contrastive learning to achieve high-precision inference on sparse and few-shot single-cell RNA-seq data; and GeneLink+ [16], which combines residual-GATv2 [17] with an improved dot-product mechanism to alleviate challenges of data sparsity, cellular heterogeneity, and over-smoothing.

However, existing GRN inference methods still face critical challenges: **fixed GNN architectures struggle to adapt to the variability in gene expression profiles across different scRNA-seq datasets, resulting in degraded GRN inference performance**. To intuitively observe the performance benefits of adapting the variability in gene expression profiles using different architecture components, we conduct an experiment in which the state-of-the-art method GMFGRN [2] is executed with manually adjusted architecture components. Fig. 1 shows the performance of GNNs under different convolution types and fusion strategies. The results indicate that optimal performance on different datasets may require different convolution operations or fusion mechanisms. Therefore, it is necessary to design dataset-specific GNN architectures to improve the accuracy of GRN inference. However, manually designing appropriate GNN architectures for each dataset is both time-consuming and heavily reliant on expert knowledge.

To address these challenges, we propose AutoGRN, an automated graph neural network framework for GRN inference. Unlike traditional methods relying on expert-designed GNN architectures, AutoGRN constructs a search space comprising convolution functions, activation functions, and fusion strategies tailored for GRN inference tasks. Through an automated search process, AutoGRN adapts to the characteristics of gene expression profiles in diverse scRNA-seq datasets, identifying the optimal GNN architecture for each dataset within the defined search space.

Our contributions are summarized as:

- We propose AutoGRN, an adaptive framework designed for GRN inference. To the best of our knowledge, it represents the first study to automatically design GNNs for the task of inferring GRNs from scRNA-seq datasets.
- To accelerate the search for high-performing GNN architectures, we design an entropy-constrained genetic algorithm that enhances convergence efficiency and achieves better performance within shorter search times.
- We conducted extensive experiments on six publicly available scRNA-seq datasets. The experimental results demonstrate that AutoGRN achieves state-of-the-art performance across all datasets compared to existing GRN inference methods.

II. Methodology

A. Overview of AutoGRN

To facilitate a more intuitive understanding of the proposed AutoGRN framework for automated GNN design in GRN inference, we elaborate on the overall workflow of AutoGRN. The GRN inference task based on AutoGRN consists of two stages.

In Stage 1, as illustrated in Fig. 2, we first define a comprehensive search space composed of various GNN components that are potentially influential to performance of GRN inference. A genetic algorithm with an information entropy-based constraint is then employed to generate candidate GNN architectures from this search space. Each candidate is evaluated based on its fitness for the GRN inference task. Through iterative evolution, this process aims to automatically identify the optimal GNN architecture for GRN inference. The detailed procedure of the search algorithm and information entropy-based constraints is depicted in Fig. 3.

In Stage 2, the top-K GNN architectures, identified in Stage 1 based on their task fitness, are subjected to a full training, validation, and testing process. The architecture achieving the best performance is ultimately selected as the optimal model for GRN inference.

B. Gene Regulatory Network Inference Based on the GNN Modeling Paradigm

Gene regulatory network describes the interactions between TFs and regulated/target genes, where nodes are TFs or genes, and edges are their regulatory relationships. Thus, the gene regulatory network inference task in this work is regarded as a classical link prediction problem by the GNN modeling paradigm. Link prediction aims to learn a mapping function that encodes nodes (i.e., TFs or genes) into embeddings that preserve intrinsic properties [18], and then uses the node embeddings to estimate the likelihood of interaction of TF-gene pairs based on score functions, as shown in Fig. 4.

Given a scRNA-seq matrix (i.e., bipartite graph G_b), and an adjacent matrix (i.e., graph G), the GNN modeling paradigm expects to learn a mapping function θ that can encode nodes into embeddings $\boldsymbol{e}$ to predict latent TF-gene interactions:

$$\boldsymbol{e} = \mathrm{GNN}_\theta(G_b, G). \tag{1}$$

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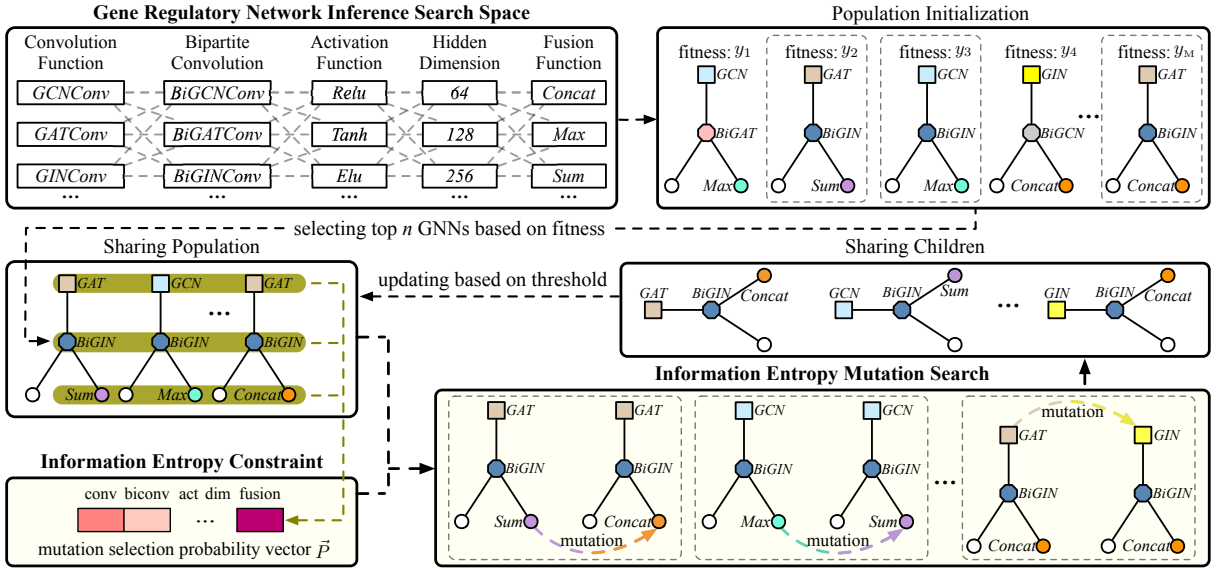


Fig. 3: An overview of AutoGRN's evolutionary search algorithm, where an information entropy-based constraint is applied to guide mutation by identifying uncertain gene positions, thus enhancing population convergence.

Based on the message passing mechanism, the GNNs in this work will iteratively update the embedding of TF/gene using its embedding and the embedding of cells in the bipartite graph G_b. After obtaining the updated embeddings, the embeddings $\boldsymbol{e}_i$ of TFs and the embeddings $\boldsymbol{e}_j$ of genes will be fused, and then the fused embeddings will be input into the MLP to estimate the interaction score s_k of TF-gene pairs:

$$s_k = \text{MLP}(\text{Fusion}(\boldsymbol{e}_i, \boldsymbol{e}_j)). \tag{2}$$

The binary cross-entropy loss function $\mathcal{L}_{BCE}$ provided in Equation 3 will be used to calculate the loss of all TF-gene pairs, as shown below:

$$\mathcal{L}_{BCE} = -\sum_{k=1}^{K} y_k \log(s_k) + (1 - y_k)\log(1 - s_k), \tag{3}$$

where k represents the k-th TF-gene pair, and y_k is the label for the k-th TF-gene pair.

C. Automated Graph Neural Network Design

In this section, we first introduce the problem formulation for automatically designing the optimal GNN model. Then, we describe the complete search space for the gene regulatory network inference task. Finally, the genetic search algorithm with information entropy constraint is presented.

Problem Definition. Given the search space S including various search components, a graph data set $\mathcal{G}$ containing a training set $\mathcal{G}_{train}$ and a validation set $\mathcal{G}_{val}$, a performance evaluation strategy E. The target of GNAS is to design the optimal GNN model g_{opt} from the search space S, where the GNN model g is trained on set $\mathcal{G}_{train}$ and obtains the best performance on set $\mathcal{G}_{val}$. The above process can be expressed as follows:

$$\begin{aligned} g_{opt} &= \text{argmax}_{(g \in S)}\, E(g_{\theta^*}, \mathcal{G}_{val}), \\ s.t. \quad \theta^* &= \text{argmin}_{\theta}\, \mathcal{L}_{BCE}(g_\theta, \mathcal{G}_{train}). \end{aligned} \tag{4}$$

Search Space. Considering the applicability to the task of GRN inference, we incorporate a series of search components suitable for this task into the search space, as shown in Fig. 3. The search components and their candidate operations are listed as follows:

Convolution Function. This component decides how nodes aggregate neighbor information. Since the features of various neighbors have differing impacts on the central node, graph convolution first computes the coefficients between the central

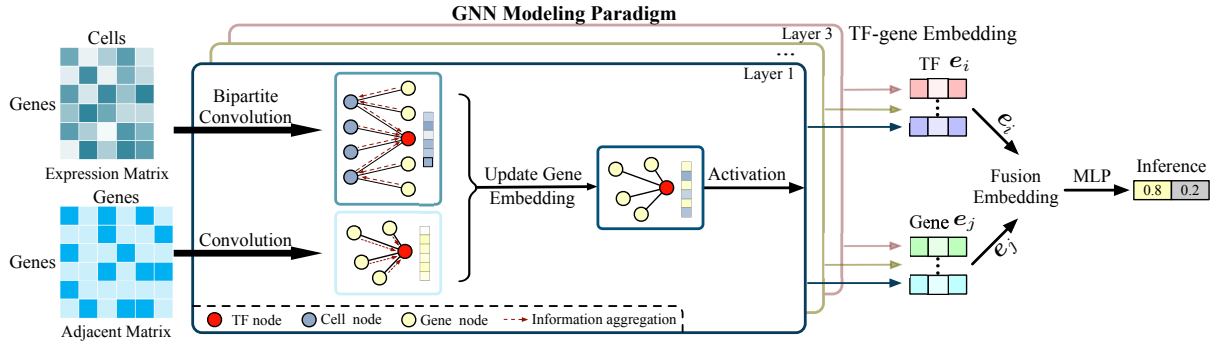


Fig. 4: Gene regulatory network inference based on the GNN modeling paradigm.

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node and its neighbors. These coefficients are then used to combine the neighbors' features, producing the central node's representation. In this paper, we implement nine types of graph convolution operations. The available convolution operations include: *SAGEConv* [19], *SAGEv2Conv* [19] and *GCNConv* [20], *GATConv* [21], *GINConv* [22], *GraphConv* [23], *GATv2Conv* [17], *TAGConv* [24], and *EdgeConv* [25].

Bipartite Convolution.

In the GRN inference task, the scRNA-seq data naturally forms a gene-cell bipartite graph and thus the interaction information between genes and cells is stored in a bipartite graph. Modeling gene-cell interactions explicitly allows the model to capture cell-specific regulatory dynamics and avoid confounding indirect connections. AutoGRN employs bipartite graph convolution to extract the interaction details of genes and cells. Additionally, we add *BiNoneConv* operation, which means no bipartite convolution operation in the sampled GNN model. We implement different bipartite convolution operations, including *BiSAGEConv*, *BiSAGEv2Conv*, *BiGCNConv*, *BiGATConv*, *BiGINConv*, *BiGraphConv*, *BiNoneConv*.

Activation Function. Activation functions endow the model with nonlinear fitting capabilities and play a key role in smoothing hidden representations. In this paper, the activation functions used include: *elu*, *tanh*, *relu*, *relu6*, *leaky_relu*, *hardtanh*, *celu*, *rrelu*.

Hidden Dimension. Transforming and reducing the original feature's dimension is a practical approach to improving the hidden representation through a learnable parameter θ. In this paper, the hidden dimensions are chosen from 64, 128, 256, 512, and 1024.

Fusion Function. In GRNs, the existence of regulatory relationships between TFs and genes is usually determined by their biological functions, regulatory mechanisms, and synergistic effects. Examining the embedding of TF or gene alone cannot fully reveal their regulatory relationship. Through embedding fusion, it is possible to comprehensively consider the regulatory signals between TF-gene pairs, to better model their complex interactions. In this work, we provide five approaches to fuse embeddings of TF-gene pairs: *concat*, *multiply*, *sum*, *max*, and *abs_difference*.

Search Algorithm. As depicted in Fig. 3, to efficiently identify the optimal GNN model, we utilize a genetic search algorithm with information entropy constraint, which leverages valuable information to constrain the search direction, thereby accelerating the search process. The search process of AutoGRN is optimized as follows:

1) Population Initialization. Randomly sampling M GNN models from the search space as the initial population. With loss function $\mathcal{L}_{BCE}$, these GNN models are trained based on gradient descent using the training set $\mathcal{G}_{train}$. The fitness of these GNN models is then obtained based on the performance evaluation strategy, which in this work is the AUROC on the validation set $\mathcal{G}_{val}$;

2) Sharing Population. Ranking the initial population and selecting top n GNN models as the sharing population according to the evaluated fitness;

3) Information Entropy Constraint. Calculating the frequency distribution of the candidate operations of each search component in the sharing population to derive the information entropy vector $\vec{H} = [h(c_1), h(c_2), ..., h(c_5)]$:

$$h(c_i) = -\sum_{j=1}^{n} f(v_i^j) \log_2 f(v_i^j), \tag{5}$$

where c_i represents the i-th search component, v_i^j denotes the j-th candidate operation of the i-th component, and $f(v_i^j)$ is the frequency of v_i^j in the i-th component.

The information entropy vector $\vec{H}$ is then normalized using the softmax function to obtain the mutation selection probability vector $\vec{P} = [p_1, p_2, ..., p_5]$:

$$p_i = \frac{e^{h(c_i)}}{\sum_{j=1}^{5} e^{h(c_j)}}. \tag{6}$$

4) Information Entropy Mutation Search. Selecting n parent GNN models from the sharing population based on the evaluated fitness. If p_i is higher, it represents that the candidate operations in the i-th search component are more diverse in sharing population and require more mutation. Thus, when parent GNN models are mutated to generate n new child GNN models, the soft sampling is used to enable sufficient mutation of the i-th search component and keep the diversity of GNN models. In the soft sampling, the mutation selection probability of search components of each GNN model is set to the mutation selection probability vector $\vec{P}$;

5) Sharing Children & Updating. Gathering the generated child GNN models as the sharing children. One of the sharing children is added to the sharing population if its fitness is greater than the threshold, such as the average fitness of the sharing population.

When the search process is completed, AutoGRN can obtain the optimal GNN model by re-training top k GNN models in the sharing population.

III. Experiments

A. Datasets and Preprocessing

We next evaluate the performance of AutoGRN on several datasets. The following datasets were used in our evaluation: 1) bone marrow-derived macrophages (**boneMarrow**) [26]; 2) IB10 mouse embryonic stem cells (**mESC(1)**) [27]; 3) 5G6GR mouse embryonic stem cells (**mESC(2)**) [28]; 4) mouse hematopoietic stem cells with erythroid-lineage (**mHSC(E)**) [29]; 5) mouse hematopoietic stem cells with granulocyte-monocyte-lineage (**mHSC(GM)**) [2]; and 6) mouse hematopoietic stem cells with lymphoid-lineage

Cell line	#Genes	#Cells	#Samples	#TFs
BoneMarrow	20,463	6,283	50,254	13
mESC(1)	24,175	2,717	154,931	38
mESC(2)	18,385	421	150,541	18
mHSC(E)	4,762	1,071	44,735	18
mHSC(GM)	4,762	889	44,741	18
mHSC(L)	4,762	847	44,738	18

TABLE I: Statistical summary of the six scRNA-seq datasets used in this paper.

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(**mHSC(L)**) [2]. The embeddings used in our experiments were obtained from the initial preprocessing of the original datasets, as performed by GMFGRN [2]. A statistical summary of each dataset is provided in Table I.

B. Experimental Settings

The experimental procedure comprises two stages: GNN architecture search and model evaluation.

1) **GNN Architecture Search Settings:** As shown in Fig. 6, we analyzed the impact of hyperparameter configurations on the performance of GRN inference. To balance performance improvement and computational efficiency, we configured the genetic algorithm-based architecture search with an initial population size of 100 and limited the search process to six iterations. In each epoch, candidate architectures were evaluated for fitness based on their performance on the validation set. Using roulette wheel selection, 10 individuals were selected to form the parental population based on their fitness scores. This strategy effectively controlled computational resource consumption while preserving diversity within the search space. In addition, an entropy-constrained algorithm was employed to dynamically adjust the mutation probability of each genome, thereby accelerating convergence and enhancing exploration.

2) **GNN Model Evaluation Settings:** After the entropy-constrained genetic algorithm completed, the optimal GNN architecture for the target dataset was selected. AutoGRN then performed comprehensive evaluations using this architecture. During testing, the model was retrained from scratch without any manual hyperparameter tuning, ensuring an unbiased evaluation of its performance.

C. Baselines

We compared the proposed AutoGRN with eight state-of-the-art baseline methods, including information theory-based, deep learning-based, and GNN-based approaches. The information theory-based baselines include PCC, Normi [30], and MI. The deep learning-based baselines include CNNC [12], MetaSEM [31], STGRNS [13], and DeepDRIM [32]. The GNN-based baselines include GMFGRN [2], scMGATGRN [14] and GeneLink+ [16]. All baselines were evaluated using three-fold cross-validation, ensuring that the training and testing sets did not share any TFs.

- **PCC** assumes a linear relationship between gene co-expression strength and the likelihood of regulatory interactions, and employs the Pearson correlation coefficient to infer GRNs from gene expression data.
- **Normi** integrates gene expression normalization with adaptive gene module detection, enabling robust and biologically meaningful gene regulatory network inference across heterogeneous single-cell datasets.
- **MI** utilizes mutual information between gene expression profiles to capture complex, non-linear dependencies beyond simple correlations.
- **CNNC** leverages deep learning to encode gene expression data, integrating co-expression analysis and causal inference into a unified framework, while enhancing the interpretability and scalability of GRN inference.
- **MetaSEM** introduces a meta-learning-based framework for GRN inference, combining structural equation modeling (SEM) with a bi-level optimization strategy to identify key regulators under high-dimensional sparse data with limited labels, thereby improving robustness when dealing with small-scale datasets.
- **STGRNS** uses a Transformer-based framework with gene expression motifs to enhance GRN inference from scRNA-seq data.
- **DeepDRIM** utilizes a deep neural network with primary and neighbor image embeddings to reconstruct cell-type-specific gene regulatory networks from single-cell RNA-seq data, effectively eliminating false positives caused by transitive interactions.

Dataset	Wilcoxon p		t-test p	
	AUROC	AUPRC	AUROC	AUPRC
boneMarrow	1.95e-03	1.95e-03	1.42e-03	2.60e-04
mESC(1)	1.95e-03	1.37e-02	1.56e-03	9.93e-03
mESC(2)	9.77e-03	1.95e-03	6.96e-03	8.00e-04
mHSC(E)	1.95e-03	4.45e-02	5.10e-04	3.97e-02
mHSC(GM)	1.95e-03	3.91e-03	7.96e-07	1.34e-03
mHSC(L)	1.95e-03	3.91e-03	4.26e-07	1.24e-03

TABLE II: Statistical significance test results of AUROC and AUPRC scores between AutoGRN and the state-of-the-art baseline model GMFGRN on six datasets, using the Wilcoxon signed-rank test and paired t-test.

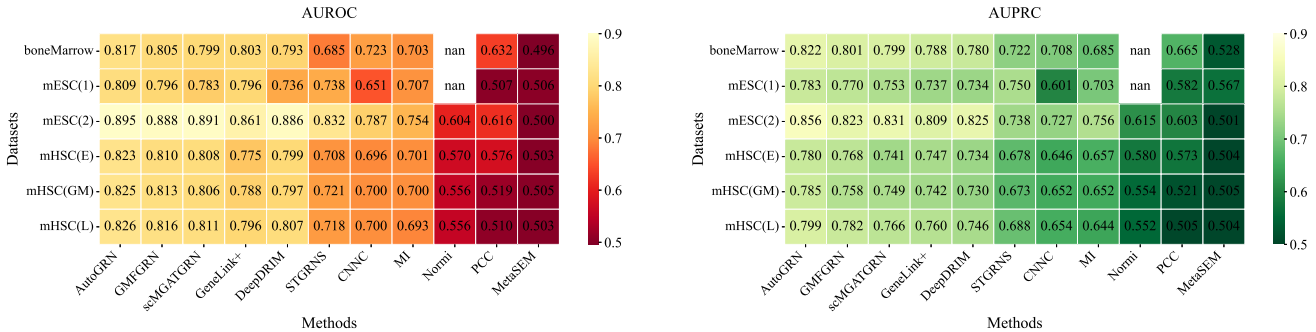


Fig. 5: Comparison of AutoGRN with state-of-the-art baselines on six datasets. The symbol 'nan' indicates that it took the Normi method too long to run on these datasets, and its results could not be obtained.

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- **GMFGRN** integrates matrix factorization with graph neural networks to learn gene embeddings, enabling efficient and accurate inference of potential target genes in GRNs with reduced computational cost.
- **scMGATGRN** leverages multiview graph attention and view-level attention to integrate local and high-order neighbor features, enhancing GRN inference from scRNA-seq data.
- **GeneLink+** introduces a residual-GATv2-based framework that combines dynamic attention and learnable weighted dot products to accurately and interpretable infer causal regulatory interactions from diverse transcriptomic data.

D. Performance Comparison

Fig. 5 summarizes the overall performance comparison between AutoGRN and the baseline methods on GRN inference tasks across six representative benchmark datasets. The results show that AutoGRN consistently outperforms competing methods to varying degrees in terms of both AUROC and AUPRC metrics. To assess the statistical significance of the observed differences, we conducted paired t-tests and Wilcoxon signed-rank tests on performance metrics obtained from multiple random seeds across six datasets. The resulting p-values are reported in Table II. For all datasets, both AUROC and AUPRC improvements achieved by AutoGRN are statistically significant (with $p < 0.05$), and several results even reach the $p < 0.001$ level, indicating strong evidence of superiority. Notably, the fixed-structure GNN method GMFGRN exhibits a slightly lower AUPRC score than the non-GNN deep learning model DeepDRIM on the mESC(2) dataset. However, with our proposed method, the prediction performance significantly exceeds that of DeepDRIM by 3.7%, further demonstrating the superiority of AutoGRN.

E. Ablation Studies

Ablation Study on the Search Space. To validate the effectiveness of the GNN search space specifically designed for the GRN inference task, we conducted comprehensive experiments by fixing a specific component within the search space while allowing other components to be fully optimized. These experiments include: 1) AutoGRN (GATConv): the convolution function is fixed to GAT, thereby ignoring the potential impact of convolution function search; 2) AutoGRN (BiNoneConv): the bipartite convolution is set to None, thereby disabling the entire bipartite convolution search; 3) AutoGRN (ReLU): the activation function is fixed to ReLU, thereby ignoring the impact of activation function search; 4) AutoGRN (256): the hidden layer dimension is fixed at 256 to eliminate the potential performance impact from searching for alternative hidden layer dimensions; 5) AutoGRN (concat): the fusion function of gene and cell embeddings is fixed to concatenation, thereby disregarding the use of other potentially effective fusion methods. As shown in Table III, the experimental results consistently demonstrate that searching across all components yields the best performance, thus proving that the GNN search space specifically designed for the GRN inference task is highly effective.

Ablation Study on the Information Entropy-Constrained. To demonstrate the effectiveness of our information entropy-constrained search algorithm, we conducted ablation experiments on multiple datasets. As

Fixed component	Variant	boneMarrow		mESC(1)		mHSC(E)		mHSC(GM)		mHSC(L)	
		AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
Convolution function	AutoGRN (GATConv)	0.807	0.809	0.793	0.776	0.815	0.773	0.815	0.768	0.817	0.787
Bipartite convolution	AutoGRN (BiNoneConv)	0.812	0.816	0.805	0.779	0.820	0.777	0.821	0.778	0.823	0.792
Activation function	AutoGRN (ReLU)	0.811	0.815	0.806	0.781	0.818	0.776	0.821	0.777	0.820	0.791
Hidden dimension	AutoGRN (256)	0.814	0.820	0.808	0.783	0.822	0.778	0.820	0.779	0.825	0.798
Fusion function	AutoGRN (concat)	0.813	0.818	0.806	0.780	0.813	0.772	0.814	0.765	0.821	0.793
None	AutoGRN (all)	**0.817**	**0.822**	**0.809**	**0.783**	**0.823**	**0.780**	**0.825**	**0.785**	**0.826**	**0.799**

TABLE III: Ablation experiment of AutoGRN with different search spaces. The first column represents the architecture component being explored, while the second column indicates that each component is fixed to a specific candidate operation.

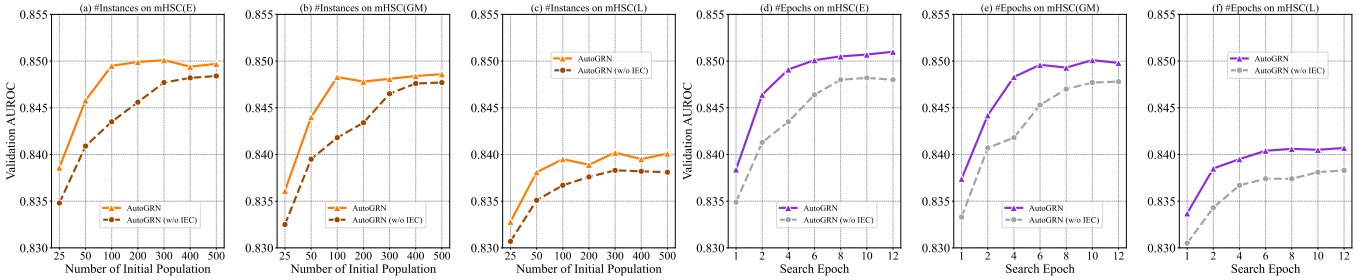


Fig. 6: Performance variation of AutoGRN before and after removing the information entropy constraint. (a), (b), and (c) show performance variation on the mHSC(E), mHSC(GM), and mHSC(L) datasets under different initial population sizes. (d), (e), and (f) illustrate performance variation on the same datasets under different numbers of search epochs. "w/o IEC" denotes the search algorithm without the information entropy constraint.

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Method	boneMarrow		mESC(1)		mESC(2)		mHSC(E)		mHSC(GM)		mHSC(L)	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
Random Search	0.799	0.798	0.801	0.774	0.873	0.805	0.801	0.751	0.813	0.761	0.800	0.732
TPE Search	0.813	0.811	0.790	0.773	**0.898**	**0.867**	0.816	0.770	0.817	0.773	0.807	0.752
AutoGRN	**0.817**	**0.822**	**0.809**	**0.783**	0.895	0.856	**0.823**	**0.780**	**0.825**	**0.785**	**0.826**	**0.799**

TABLE IV: Performance comparison of different search algorithms on six datasets. The table shows AUROC and AUPRC scores for each method across different datasets. Bold values indicate the best performance in each column.

Target gene of SP4	Validation	Target gene of BCL11A	Validation
C17ORF82	harmonizome, hTFtarget	MTRNR2L2	harmonizome
MTRNR2L2	hTFtarget	MTRNR2L8	harmonizome
MTRNR2L8	Unconfirmed	RGPD2	Unconfirmed
NDUFS7	harmonizome, hTFtarget	ZER1	harmonizome
ISOC2	harmonizome, hTFtarget	ZNF3	harmonizome
AP2B1	harmonizome, hTFtarget	NUP54	harmonizome
HIST1H4C	harmonizome, hTFtarget	MTRNR2L6	harmonizome
ATG9A	harmonizome, hTFtarget	PRDX1	harmonizome, hTFtarget
RGPD2	Unconfirmed	MTRNR2L10	harmonizome
PLS1	harmonizome, hTFtarget	GSTP1	harmonizome
BTRC	harmonizome, hTFtarget	IGF2BP3	harmonizome, hTFtarget
NKIRAS1	harmonizome, hTFtarget	KCNIP4	Unconfirmed
PYROXD1	harmonizome, hTFtarget	SCNN1A	harmonizome
MTMR7	harmonizome	AP2M1	Unconfirmed
UCHL1	harmonizome, hTFtarget	PAM	harmonizome
CDCA7L	harmonizome, hTFtarget	DUSP6	harmonizome, hTFtarget
SMPD3	harmonizome, hTFtarget	RNF219	harmonizome
MCOLN1	harmonizome, hTFtarget	AXL	harmonizome
ZER1	harmonizome, hTFtarget	H2AFY	harmonizome
ZFAND4	harmonizome, hTFtarget	RFX5	harmonizome

TABLE V: AutoGRN-predicted top 20 target genes of SP4 and BCL11A on the hESC dataset and their validation results in the Harmonizome and hTFtarget databases.

shown in Fig. 6, the experimental results indicate that the variant without entropy constraints (AutoGRN w/o IEC) requires more search epochs and a larger initial population size to converge compared to AutoGRN. Furthermore, we observe that as the number of search epochs increases and the results gradually converge, the performance gap becomes more pronounced for different initial population sizes. This highlights the effectiveness of the proposed information entropy-constrained search algorithm.

Ablation Study on the Search Algorithm. To evaluate the effectiveness and robustness of our proposed entropy-constrained genetic search algorithm, we conducted an ablation study by replacing it with two widely used alternatives: Random Search and Tree-structured Parzen Estimator (TPE) based on Bayesian optimization. For fairness, all search algorithms were run with the same number of trials. As shown in Table IV, AutoGRN achieves the highest AUROC and AUPRC on five out of six datasets and matches the best AUROC on the remaining two. This demonstrates its ability to consistently find high-quality architectures with minimal performance variation.

F. Case Study

To evaluate AutoGRN's ability to predict regulatory relationships for previously unseen TFs, we conducted a case study on the human embryonic stem cell (hESC) dataset. First, we applied 3-fold cross-validation, splitting the 18 known TFs into three subsets. In each fold, two subsets (12 TFs) were used for training, and the remaining subset (6 TFs) was held out for testing. Importantly, TFs in the test set did not appear in the training set, ensuring a strict evaluation on unseen TFs. For each fold, AutoGRN was trained on TF-Gene interaction data corresponding to the 12 training TFs, and then used to predict target genes for the 6 unseen TFs in the test set. Predicted interactions were ranked by confidence scores, and the top 100 edges were selected to construct a local regulatory subnetwork, as shown in Fig. 7.

In this GRN, node size reflects out-degree; TFs are shown in red and target genes in blue. Edges are color-coded: green for true positive regulations and orange for false positives. Among the 100 edges, 83 are green and 17 are orange. Based on the inferred GRN, we observed that the transcription factors SP4 and BCL11A play crucial regulatory roles: SP4 exhibits regulatory relationships with 40 genes in this local GRN, whereas BCL11A regulates 24 genes.

To verify the accuracy of the predicted potential target genes for the transcription factors SP4 and BCL11A, we manually validated the top 20 predicted target genes (ranked by their predicted scores) for each using the Harmonizome [33] and hTFtarget [34] databases. As shown in Table V, for the SP4 transcription factor, 18 out of 20 predicted edges were confirmed, yielding a validation rate of 90.00%. For the BCL11A transcription factor, 17 out of 20 predicted edges were validated, corresponding to a validation rate of 85.00%. In total, out of 40 predicted edges, 35 were confirmed, resulting in an overall validation rate of 87.50%. These results demonstrate the robustness and biological relevance of AutoGRN's predictions, highlighting its potential for discovering novel

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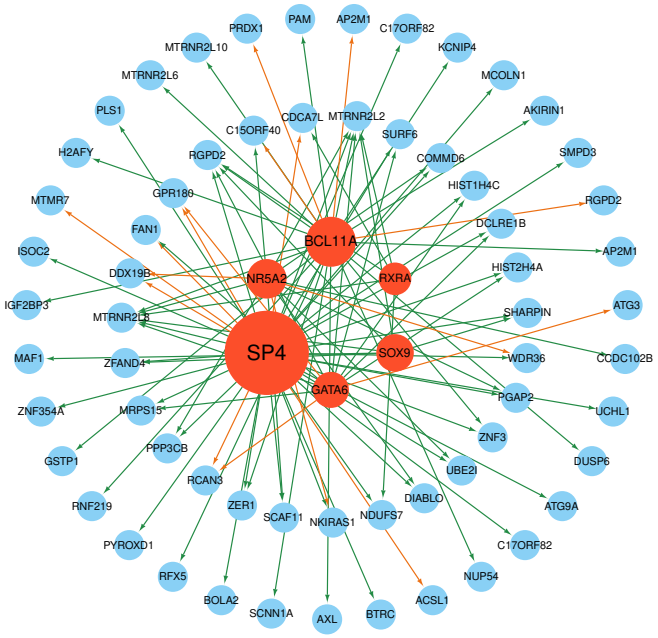
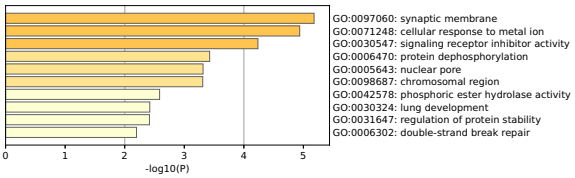
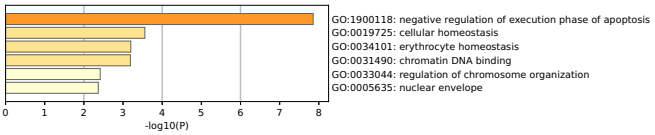


Fig. 7: The visualization of the gene regulatory network predicted by AutoGRN on the hESC dataset.



(a) SP4 Transcription Factor



(b) BCL11A Transcription Factor

Fig. 8: The results of the GO enrichment on the hESC dataset. The input for GO enrichment is AutoGRN-predicted top 50 target genes of SP4 (a) and BCL11A (b).

regulatory interactions in complex gene regulatory networks.

G. Functional Enrichment Analysis

To validate the biological plausibility of the inferred GRN, we conduct a functional enrichment analysis to reveal the known regulatory relationships. The functional enrichment analysis takes the AutoGRN-predicted top 50 target genes of transcription factors SP4 and BCL11A on the hESC dataset as input. Based on Metascape [35] with default parameters, the functional enrichment analysis in the biological processes, molecular functions, and cellular components is carried out, as shown in Fig. 8.

First, relevant studies indicate that SP4 is not only a high-risk gene for psychiatric disorders such as bipolar disorder and schizophrenia but may also serve as a network hub regulating many other schizophrenia risk genes [36]. By enriching the target genes of SP4 (Fig. 8a), Gene Ontology (GO) terms are related to schizophrenia, i.e., the known regulatory relationships based on SP4, such as synaptic membrane (GO:0097060) [37], and signaling receptor inhibitor activity (GO:0030547) [38]. In addition, the disturbances in metal ion homeostasis are associated with schizophrenia [39], validating cellular response to metal ion (GO:0071248).

Then, BCL11A is a critical transcriptional repressor of the γ-globin genes (HBG1/2) in adult erythrocytes. It inhibits the production of fetal hemoglobin (HbF) by binding to enhancer and promoter regions of the β-globin gene cluster [40]. In patients with sickle cell disease (SCD) and β-thalassemia, inhibition of BCL11A can reactivate HbF, compensating for defective adult hemoglobin (HbA) and alleviating the symptoms of SCD [41]. Thus, by enriching the target genes of BCL11A (Fig. 8b), GO terms are associated with hemoglobin, such as negative regulation of execution phase of apoptosis (GO:1900118) [42], [43], and cellular homeostasis (GO:0019725) [42], [43]. Especially, the term of the erythrocyte homeostasis (GO:0034101) enhances the enrichment explanation of the target genes of BCL11A and reveals the known regulatory relationships.

To summarize, the target genes of transcription factors are enriched in related functions, which conform to known regulatory relationships. Thus, these enrichment results indicate that the GRN inferred by AutoGRN is biologically plausible.

H. Cross-dataset Architecture Transferability Analysis

To assess the transferability of architectures discovered by AutoGRN, we conduct a cross-dataset architecture transfer experiment. Specifically, we transfer the optimal architecture searched on a source dataset to other datasets for testing. As shown in Table VI, the results overall show that the transferred optimal architectures achieve comparable performance to those searched directly on the target dataset. For example, applying the optimal architecture discovered on mHSC(GM) to predict mHSC(L) yields an AUROC of 0.824 and an AUPRC of 0.791, compared to 0.826 and 0.799 when searched directly on mHSC(L), indicating relatively small performance degradation. However, in some cases the gap is larger, such as transferring the architecture from mESC(2) to mESC(1) (AUROC: 0.787 vs. 0.809). These observations reflect that

Source Dataset	Target Dataset	AUROC	AUPRC
mESC(1)	mESC(2)	0.878 (0.895)	0.841 (0.856)
mESC(2)	mESC(1)	0.787 (0.809)	0.773 (0.783)
mHSC(E)	mHSC(GM)	0.818 (0.825)	0.771 (0.785)
mHSC(E)	mHSC(L)	0.820 (0.826)	0.787 (0.799)
mHSC(GM)	mHSC(E)	0.807 (0.823)	0.763 (0.780)
mHSC(GM)	mHSC(L)	0.824 (0.826)	0.791 (0.799)
mHSC(L)	mHSC(E)	0.815 (0.823)	0.774 (0.780)
mHSC(L)	mHSC(GM)	0.822 (0.825)	0.777 (0.785)

TABLE VI: Performance of cross-dataset architecture transferability. AutoGRN searches for the optimal architecture on a source dataset and transfers it to other target datasets for testing. The performance marked in gray denotes that the optimal architecture is searched on the target dataset.

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Adaptive Component	boneMarrow		mESC(1)		mHSC(E)	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
Convolution+Bipartite	0.815	0.818	0.805	0.779	0.816	0.776
Convolution+Activation	0.814	0.817	0.803	0.778	0.815	0.774
Convolution+Fusion	0.816	0.821	0.807	0.782	0.821	0.778
Bipartite+Activation	0.805	0.807	0.794	0.773	0.811	0.765
Bipartite+Fusion	0.812	0.815	0.801	0.777	0.819	0.776
Activation+Fusion	0.813	0.817	0.800	0.775	0.820	0.778
All	**0.817**	**0.822**	**0.809**	**0.783**	**0.823**	**0.780**

TABLE VII: Interaction effects of different search components. Bold values indicate the best performance in each column.

Datasets	Cost Time (h)		Memory (GB)	
	GMFGRN	AutoGRN	GMFGRN	AutoGRN
boneMarrow	1.25	3.05	8.63	10.60
mESC(1)	1.03	2.67	6.91	19.83
mESC(2)	0.45	1.24	5.63	17.75
mHSC(E)	0.19	0.56	2.23	9.85
mHSC(GM)	0.27	0.48	2.58	9.53
mHSC(L)	0.28	0.47	2.67	6.25

TABLE VIII: Comparison of runtime and memory usage between GMFGRN and AutoGRN across six datasets.

AutoGRN is capable of discovering cross-dataset architectures with good robustness, certain components may overfit to the graph distribution in the training data, indicating the rationality of searching for the optimal architecture for each specific dataset.

I. Interaction Effects of Different Search Components

To explore the interaction effects of different search components, we conduct a series of controlled experiments in which pairwise search components are adaptively searched while keeping others fixed. As shown in Table VII, the interaction between Convolution Function search component and Fusion Function search component is crucial, making a significant contribution to generalization performance. Moreover, the best performance is obtained when all components are jointly optimized, achieving the highest AUROC and AUPRC scores. In addition, the combination of Convolution+Bipartite is consistently superior to Convolution+Activation, indicating stronger interaction effects between similar search components. The combination of Bipartite+Activation achieves the worst adaptive results, showing weaker interaction between these two search components. In summary, by exploring the interaction effects of different search components, we can observe that component interactions do affect performance, and joint optimization of different components leads to further improvements.

J. Computational Environment and Efficiency

In this study, all experiments were conducted on a workstation running on Ubuntu 22.04, equipped with an AMD EPYC 7542 32-core CPU, an NVIDIA GeForce RTX 3090 GPU (24 GB), and 128 GB of RAM. On average, the architecture search phase of AutoGRN took approximately 1.41 hours per dataset, with GPU memory usage of 12.3 GB. To quantify the computational cost of AutoGRN, Table VIII compares the runtime and memory usage of AutoGRN with GMFGRN across six benchmark datasets. While AutoGRN incurs a moderate overhead from architecture search, it remains efficient and deployable on standard hardware, confirming its practicality for real-world use.

IV. Conclusion and Future Direction

In this study, we proposed AutoGRN, an automated graph neural network framework designed for GRN inference tasks. By constructing an effective search space, AutoGRN adaptively optimizes GNN architectures for different scRNA-seq datasets using a genetic search algorithm constrained by information entropy. Experimental results demonstrate that AutoGRN consistently outperforms existing GRN inference methods across multiple public datasets. In addition, the results of the case study and functional enrichment analysis can validate the biological plausibility of the inferred GRN predicted by AutoGRN, thus showing potential for clinical applications such as disease gene regulation analysis.

Despite achieving promising results, analyzing the limitations of AutoGRN to explore future directions is crucial. For instance, AutoGRN is currently only applicable to single-omics data and has not been extended to multi-omics settings, such as chromatin accessibility and DNA methylation. The multi-omics data integration represents a growing trend in GRN inference research. For cross-modal omics data, the heterogeneous graph convolution function [44] can encode embeddings from different modalities into a shared latent space. In future work, we plan to incorporate a search component containing different heterogeneous graph convolution functions in the search space to enhance the ability of cross-modal. However, cross-modal omics data are usually measured independently [45], resulting in unpaired data. Hence, to integrate cross-modal omics data with the heterogeneous graph convolution function, the alignment methods (such as GLUE [45], Seurat [46], and bindSC [47]) are used for initial pairing. Naturally, a search component containing different alignment methods will be incorporated into the search space to ensure the correct pairing of multi-omics data. This extension will enable AutoGRN to model complex regulatory mechanisms from diverse perspectives, laying the foundation for more clinically relevant GRN inference.

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